

# From your first qubit to block encoding

A coherent path through the lectures you need,  
with the mathematics explained as it appears.

## The question this guide answers

How can a quantum circuit use a matrix that is not itself a quantum gate?

$$A = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \longrightarrow U = \begin{pmatrix} A/\alpha & B \\ C & D \end{pmatrix}$$

You will learn to read a quantum state, follow a circuit, measure an auxiliary qubit, and derive a block encoding yourself. One small matrix stays with us throughout the construction.

Selected-topic companion to the  
2026 Gene Golub SIAM Summer School at Duke

Prepared 16 September 2026. Original explanations, worked calculations, and vector diagrams; linked primary sources throughout.

## How to use this guide

Read one lesson, redo its calculations with a pencil, and then watch the corresponding lecture. A lecture may contain more than you need for this goal; the table below identifies the relevant material. The exercises have worked answers near the end.

This is an independently written tutorial organized around the program you supplied. It is **not a transcript or a claim to have reviewed the full recordings**. The research comes from public course pages, mathematical lecture notes, and the primary references listed at the end. The prose uses an example-first teaching approach; it does not copy any author's distinctive wording.

| Step | Program item                                      | What you need to carry forward                                        | Page               |
|------|---------------------------------------------------|-----------------------------------------------------------------------|--------------------|
| 1    | Day 1, Lecture 1: Di Fang                         | States, amplitudes, vectors, gates.                                   | <a href="#">3</a>  |
| 2    | Day 1, Lecture 2: Di Fang                         | Unitarity, circuits, measurements; the meaning of a controlled gate.  | <a href="#">5</a>  |
| 3    | Day 1, Lecture 3: Di Fang                         | Joint states and partial measurement.                                 | <a href="#">8</a>  |
| 4    | Day 1, Lecture 4: Di Fang                         | Reversible computation, workspace, uncomputation, access assumptions. | <a href="#">10</a> |
| 5    | Day 2, Tutorial 1: Daniel Nino; Hadamard exercise | Interference between two operations.                                  | <a href="#">12</a> |
| 6    | Day 4, Lecture 1: Di Fang                         | The block-encoding definition and a simple construction.              | <a href="#">14</a> |
| 7    | Day 4, Lecture 2: Di Fang                         | General linear combination of unitaries (LCU).                        | <a href="#">17</a> |
| 8    | Day 4, Tutorial 3; LCU exercise                   | Calculate and verify the whole circuit.                               | <a href="#">20</a> |
| 9    | Day 5, Lecture 3: Chao Yang                       | Use matrix structure to build a useful encoding.                      | <a href="#">22</a> |

**Afterward:** the final lesson introduces just enough singular values and amplitude amplification to orient you in the rest of Day 4 Lecture 2 and Day 5 Lecture 4. Their full proofs are a second course.

**Deferred topics:** teleportation, the detailed SWAP test, Deutsch–Jozsa, Bernstein–Vazirani, QFT, quantum phase estimation, and Trotterization are not needed to derive the constructions in this guide. We briefly connect some of them to the story so the omissions are clear.

**Source connection.** The lecture mapping follows the supplied Duke program. Use its recording links when watching; this PDF does not reproduce the summer school's restricted slides, recordings, or worksheets. [Open reference](#).

# 1. A qubit is a vector you can act on

## DAY 1, LECTURE 1

*Start with ordinary numbers. The new quantum vocabulary will be names for precise mathematical objects.*

### A vector is an ordered list

A column vector is a list written vertically. The order matters: the first slot and second slot mean different things. For a qubit, we label the slots with the possible measurement results, 0 and 1:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The notation  $|0\rangle$  is pronounced “ket zero.” These two vectors are called the **computational basis**. They act like two coordinate directions: any two-entry vector can be built from them.

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}.$$

The letter  $\psi$  (psi) is just a name. The numbers  $a$  and  $b$  are **amplitudes**. Saying that a state is a superposition means writing it as a linear combination such as this one. It does not mean that a measurement hands you both entries.

### Amplitudes become probabilities by squaring their magnitudes

For a real number, its magnitude is its absolute value. A valid pure qubit state satisfies

$$|a|^2 + |b|^2 = 1.$$

Measuring in the computational basis returns 0 with probability  $|a|^2$  and 1 with probability  $|b|^2$ . This is the Born rule. The sum is one because some outcome must occur. Our first example is

$$|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle.$$

It is normalized:  $9/25 + 16/25 = 1$ . Over many independently prepared runs, about 36% of measurements give 0 and 64% give 1. A single run gives one bit, not the pair  $(3/5, 4/5)$ .

### The length of a vector

The **norm** of  $v = (v_0, v_1)^T$  is  $\|v\| = \sqrt{|v_0|^2 + |v_1|^2}$ . Thus a state vector has norm one. If a nonzero calculation produces  $v = (3, 4)^T$ , dividing by its norm, 5, gives a state vector  $(3/5, 4/5)^T$ . This process is called **normalization**.

#### Carry this forward

Keep amplitudes and probabilities separate. We add and transform amplitudes during a circuit; we square magnitudes when predicting a measurement.

**Source connection.** IBM’s single-system lesson supplies the state-vector and measurement framework. The numerical calculations here and below are worked out for this guide. [Open reference](#).

## Why signs and complex numbers matter

A quantum amplitude can be negative or complex. A complex number has the form  $z = x + iy$ , where  $i^2 = -1$ . Its conjugate is  $z^* = x - iy$ , and

$$|z|^2 = z^*z = x^2 + y^2.$$

For example,  $|3i/5|^2 = 9/25$ . The state  $(3i/5)|0\rangle + (4/5)|1\rangle$  therefore has the same computational-basis probabilities as our previous example. The phase information can nevertheless affect what happens after a gate.

## A matrix is a linear transformation

For now, “linear” means that the same rule applies to each component and respects addition:

$$M(a|0\rangle + b|1\rangle) = aM|0\rangle + bM|1\rangle.$$

Matrix-vector multiplication is a concrete recipe:

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} pa + qb \\ ra + sb \end{pmatrix}.$$

Each row gives one new entry. Equivalently, the two columns tell you where  $|0\rangle$  and  $|1\rangle$  go. The  $X$  gate exchanges them:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X\left(\frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle\right) = \frac{4}{5}|0\rangle + \frac{3}{5}|1\rangle.$$

The  $Z$  gate changes one sign:  $Z = \text{diag}(1, -1)$ . Here  $\text{diag}$  means a diagonal matrix with the listed entries and zeros elsewhere.

The Hadamard gate  $H$  mixes the entries by taking a sum and a difference:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H \begin{pmatrix} a \\ b \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} a + b \\ a - b \end{pmatrix}.$$

For our state, the result is  $(7|0\rangle - |1\rangle)/(5\sqrt{2})$ . Its measurement probabilities are  $49/50$  and  $1/50$ . Changing the input to  $(3|0\rangle - 4|1\rangle)/5$  instead gives  $(-|0\rangle + 7|1\rangle)/(5\sqrt{2})$ , reversing those probabilities. A sign that was invisible in an immediate measurement becomes visible after  $H$ .

### This is interference

Contributions to the same output amplitude can reinforce or cancel. In the first calculation,  $3/5 + 4/5$  reinforces and  $3/5 - 4/5$  partly cancels. Block encoding will use exactly this kind of arithmetic.

### Pause and try 1

Apply  $X$  and then  $H$  to  $|0\rangle$ . What are the final amplitudes and measurement probabilities?

**Source connection.** For a slower introduction to multiplication and inverses, see Gilbert Strang’s MIT course session. [Open reference.](#)

## 2. Which matrices are quantum gates?

### DAY 1, LECTURE 2

We can multiply a vector by any compatible matrix. A quantum gate has an additional obligation: it must preserve the state's length.

#### Unitarity is the rule for the full operation

In the ideal circuit model, a gate on a closed system is represented by a **unitary** matrix  $U$ . Let  $U^\dagger$  denote its conjugate transpose: exchange rows with columns and conjugate each entry. Unitarity means

$$U^\dagger U = U U^\dagger = \mathbb{I}.$$

The identity  $\mathbb{I}$  leaves a vector unchanged. Consequently,  $U^\dagger$  is the operation that reverses  $U$ . A unitary preserves the norm of every input, so its output probabilities still sum to one. Measurement is a different kind of operation; this rule does not say that every physical process is a unitary on the subsystem you happen to observe.

You can check that  $X^2 = Z^2 = H^2 = \mathbb{I}$ . These gates undo themselves. Other gates have a different inverse. We will often prepare something using  $P$  and reverse it using  $P^\dagger$ .

#### The matrix that will accompany us

Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

It leaves  $|0\rangle$  alone but maps  $|1\rangle$  to  $\frac{1}{2}|1\rangle$ , which has norm  $1/2$ . Also,  $A^\dagger A = \text{diag}(1, 1/4) \neq \mathbb{I}$ . Therefore  $A$  is not a gate on a single qubit.

Could we apply  $A$  and simply normalize afterward? For an input  $|\psi\rangle$ , this would prescribe  $A|\psi\rangle/\|A|\psi\rangle\|$ . The denominator depends on the input, so this is generally not a linear transformation. It cannot be a fixed deterministic unitary gate. Later, we will obtain exactly this normalized state *conditional on a measurement outcome*. The probability of that outcome accounts for the changing norm.

#### A circuit is a sequence of operations

Circuit pictures run left to right. Matrix products act on a vector right to left. Applying  $X$  first and  $H$  second is therefore  $HX|\psi\rangle$ . Reversing the order usually changes the answer. Here  $HX|0\rangle = |-\rangle$ , whereas  $XH|0\rangle = |+\rangle$ , where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

**Source connection.** IBM's circuit lesson explains the circuit notation and its relationship to matrix operations. [Open reference.](#)

## Two qubits: keeping the labels straight

A block encoding needs extra qubits, so we must learn to describe a joint system. Two qubits have four computational-basis labels:

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle.$$

Throughout this guide, the left digit is the **ancilla**, or auxiliary qubit, and the right digit is the **system** qubit containing the input we care about. This is our matrix ordering too. Software packages may use different conventions.

To combine two separately specified states, multiply their amplitudes. This operation is the **tensor product**, written  $\otimes$ :

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle.$$

For example,  $|+\rangle \otimes |0\rangle = (|00\rangle + |10\rangle)/\sqrt{2}$ . We often omit the  $\otimes$  between kets. The expression  $H \otimes \mathbb{I}$  applies  $H$  to the ancilla and leaves the system alone. A register is simply a named collection of qubits. An  $n$ -qubit register has  $2^n$  basis labels.

## A controlled gate is a coherent conditional operation

Controlled- $V$  applies  $V$  to the system when the ancilla is 1 and applies the identity when it is 0:

$$|0\rangle|\psi\rangle \mapsto |0\rangle|\psi\rangle, \quad |1\rangle|\psi\rangle \mapsto |1\rangle V|\psi\rangle.$$

There is no measurement of the control. By linearity, a superposition of control values produces a superposition of these two possibilities. In our basis ordering, its matrix is  $\text{diag}(\mathbb{I}, V)$ , with two matrix blocks.



Time runs left to right. The dot is a control, not a measurement.

With  $V = X$ , this gate is CNOT. On input  $|+\rangle|0\rangle$  it produces  $(|00\rangle + |11\rangle)/\sqrt{2}$ , a Bell state. It cannot be written as a product of two separate qubit states; this is **entanglement**. Keeping the joint state is essential when the qubits become correlated this way.

**Source connection.** IBM's multiple-system lesson develops tensor products, entangled states, and controlled operations. [Open reference.](#)

## Three ideas to recognize without getting sidetracked

### Global phase versus relative phase

Multiplying the *whole* state by a complex number of magnitude one does not change its physical predictions. Such a factor is a global phase. In contrast, changing only one amplitude's phase can alter interference. The two calculations with  $H$  in Lesson 1 demonstrate a relative sign's effect.

There is a subtlety when using controlled operations: replacing  $V$  with  $-V$  changes the sign of the control-1 branch only. That is a *relative* sign in the joint state. When building LCU circuits, signs and phases of the selected operations must be tracked.

### No cloning: why CNOT is not a universal copy button

CNOT sends  $|0\rangle|0\rangle$  to  $|00\rangle$  and  $|1\rangle|0\rangle$  to  $|11\rangle$ . But its action on  $|+\rangle|0\rangle$  gives a Bell state. Two independent copies of  $|+\rangle$  would instead be

$$|+\rangle|+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

These are different states. More generally, linearity prevents one fixed device from perfectly copying every unknown quantum state. You can still prepare multiple copies of a known state by running its preparation circuit again. Remember this when a later algorithm needs repeated trials.

**Source connection.** IBM's limitations lesson covers global phase and the no-cloning theorem. [Open reference.](#)

### Universal gate sets: a large box is a promise to build a circuit

A universal gate set lets us approximate arbitrary finite-dimensional unitary operations on qubits, up to the usual global-phase convention, by composing elementary gates. For example,  $H$ ,  $T = \text{diag}(1, e^{i\pi/4})$ , and CNOT form a standard universal set. Here  $e^{i\theta} = \cos \theta + i \sin \theta$  denotes a phase of magnitude one.

Universality does not mean that every large unitary has a short circuit. When we draw a box labeled PREPARE or SELECT, we have named an operation, but we have not yet paid for its implementation. The last two lessons will return to this distinction.

#### What you need from Lecture 2

Explain why  $A = \text{diag}(1, 1/2)$  is not unitary. Read a two-wire circuit, distinguish a control from a measurement, and reverse a known unitary with its dagger. The full universality proofs are optional for this goal.

#### Pause and try 2

Start with  $|0\rangle|1\rangle$ . Apply  $H$  to the ancilla and then controlled- $X$  to the system. Write the four-entry output vector in our ordering.

**Source connection.** Andrew Childs, sections 1.2–1.4 and Chapter 2, supplies the circuit, universality, and reversibility background. [Open reference.](#)

### 3. Measuring part of a state

#### DAY 1, LECTURE 3

*This is the conceptual bridge to block encoding. The full state stays normalized, while one measurement branch can contain a shorter vector.*

#### Group the amplitudes by the qubit you measure

Take the joint state

$$|\Psi\rangle = \sqrt{\frac{3}{5}}|00\rangle + \sqrt{\frac{1}{5}}|01\rangle + \sqrt{\frac{1}{5}}|11\rangle.$$

Its squared amplitudes add to one. To measure only the ancilla, group together the terms with the same first digit:

$$|\Psi\rangle = |0\rangle \underbrace{\left(\sqrt{\frac{3}{5}}|0\rangle + \sqrt{\frac{1}{5}}|1\rangle\right)}_{|v_0\rangle} + |1\rangle \underbrace{\left(\sqrt{\frac{1}{5}}|1\rangle\right)}_{|v_1\rangle}.$$

The branch vectors  $|v_0\rangle$  and  $|v_1\rangle$  need not have norm one. Their squared norms give the ancilla probabilities:

$$p(0) = \|v_0\|^2 = \frac{4}{5}, \quad p(1) = \|v_1\|^2 = \frac{1}{5}.$$

If we obtain ancilla 0, the remaining system state is the normalized vector

$$\frac{|v_0\rangle}{\sqrt{4/5}} = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

If we obtain ancilla 1, the system is  $|1\rangle$ . Measuring the ancilla has not necessarily measured the system in its own computational basis: the first outcome leaves a superposition.

#### Postselection is conditioning on a recorded outcome

If our experiment keeps only the runs with ancilla 0, we are **postselecting** that outcome. It is not a way to force a random event. We must report its probability, and we may need to prepare the input again for another attempt.

For a general normalized joint state written as

$$|\Psi\rangle = |0\rangle|v_0\rangle + |1\rangle|v_1\rangle,$$

the same rule applies: the chance of 0 is  $\|v_0\|^2$ , and the conditional system state is  $|v_0\rangle/\|v_0\|$ . If  $v_0 = 0$ , that outcome never occurs, and there is no conditional state to normalize.

#### The distinction that will do the work

A branch vector carries both a direction and a probability weight. After we condition on that branch, the direction becomes the system state; the squared length becomes the success probability.

## Projection notation: a compact way to select a branch

A **bra** is a conjugate-transposed ket. For example,  $\langle 0| = (1, 0)$  and  $\langle 1| = (0, 1)$ . Multiplying a bra by a ket gives an **inner product**, a single number:

$$\langle 0|0\rangle = 1, \quad \langle 0|1\rangle = 0, \quad \langle \phi|\psi\rangle = \phi_0^* \psi_0 + \phi_1^* \psi_1.$$

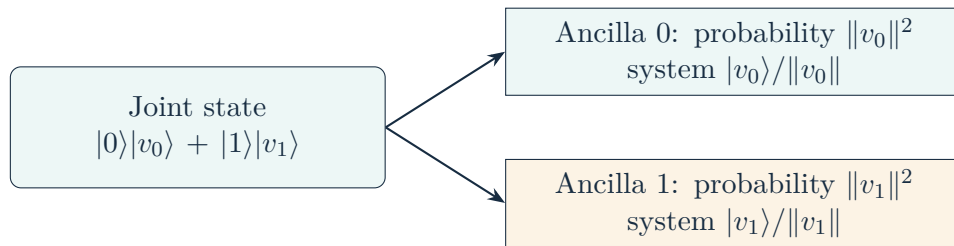
Two vectors are orthogonal when their inner product is zero. For normalized vectors, the inner product describes their overlap, including phase. The notation  $\langle 0|\otimes \mathbb{I}$  extracts the ancilla-0 branch while retaining the system:

$$(\langle 0|\otimes \mathbb{I})(|0\rangle|v_0\rangle + |1\rangle|v_1\rangle) = |v_0\rangle.$$

This is algebra describing a branch, not a standalone unitary gate. If we want to keep the joint register dimensions, use the **projector**

$$\Pi_0 = |0\rangle\langle 0|\otimes \mathbb{I}, \quad \Pi_0|\Psi\rangle = |0\rangle|v_0\rangle.$$

An outer product such as  $|0\rangle\langle 0|$  is a matrix, not a number. In this example it equals  $\text{diag}(1, 0)$ . It erases the other component and satisfies  $\Pi_0^2 = \Pi_0$ : selecting an already selected subspace does nothing further.



## What about teleportation and the SWAP test?

Teleportation combines entanglement, partial measurement, and corrections conditioned on classical results. You do not need its full protocol for block encoding. The SWAP test estimates the squared overlap of two states using a controlled exchange of their registers. We will connect it to the Hadamard test in Lesson 5.

### Pause and try 3

For  $(2|00\rangle + |01\rangle + 2|11\rangle)/3$ , find the probability of ancilla 0 and the system state conditioned on that outcome. Keep the unnormalized branch separate from the normalized answer.

**Source connection.** For the formal partial-measurement rule, see the corresponding subsection of IBM's multiple-system lesson. [Open reference.](#)

## 4. Extra workspace must be handled coherently

### DAY 1, LECTURE 4

*We now have extra qubits. We need to use them without accidentally destroying the interference we are trying to create.*

#### Reversible computation keeps enough information to undo itself

A unitary cannot send two different basis states to the same basis state. If it did, its inverse could not tell which input to recover. A classical function  $f(x)$  need not be one-to-one, so the direct map  $|x\rangle \mapsto |f(x)\rangle$  is not generally reversible.

Instead, for a one-bit output, use

$$O_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle.$$

The symbol  $\oplus$  means addition modulo 2:  $1 \oplus 1 = 0$ . The input  $x$  is retained. Applying  $O_f$  twice toggles the output twice and restores it, so this is reversible. On a superposition, it acts on every basis component by the same linear rule.

#### A tiny example of garbage blocking interference

Start with system  $|+\rangle$  and workspace  $|0\rangle$ . Copy the system's computational-basis value into the workspace using CNOT. The result is

$$\frac{1}{\sqrt{2}}(|0\rangle_{\text{sys}}|0\rangle_{\text{work}} + |1\rangle_{\text{sys}}|1\rangle_{\text{work}}).$$

Now apply  $H$  to the system. The joint result is

$$\frac{1}{2}(|0\rangle|0\rangle + |1\rangle|0\rangle + |0\rangle|1\rangle - |1\rangle|1\rangle).$$

Measuring the system gives 0 or 1 with equal probability. In contrast, without the CNOT, applying  $H$  to  $|+\rangle$  would give 0 with certainty. The workspace has retained a label distinguishing the two paths; ignoring it does not make those labels disappear.

#### Uncomputation removes the label reversibly

Before the final  $H$ , apply CNOT a second time. This restores the joint state to  $|+\rangle_{\text{sys}}|0\rangle_{\text{work}}$ . Then  $H$  produces system  $|0\rangle$  with certainty. This is the simplest example of **uncomputation**: undo temporary bookkeeping so paths can recombine.

For a larger calculation, the pattern is compute a temporary value, use it in a way that preserves the information needed to reverse the computation, and then run the computing circuit backward. It does not mean erasing arbitrary entangled workspace for free.

**Source connection.** Childs, section 1.4, discusses reversible computation. Lin, section 1.7, gives a circuit treatment of uncomputation. [Open reference](#).

## What an oracle promise does, and does not, give you

An **oracle** is a specified operation that an algorithm assumes it can call. For example, a matrix-value oracle might reversibly calculate an entry  $A_{ij}$  from its row and column labels. A *query count* counts calls to that operation. A *gate count* counts the elementary gates used to implement the algorithm, including the oracle when its implementation is known.

These are different questions. An algorithm that calls a matrix oracle ten times is not necessarily cheap if each call requires a very large circuit. Similarly, writing a state as a vector on paper does not supply a short circuit to prepare it. We will treat preparation and access costs as part of the problem, not as automatic gifts.

## A small phase-kickback bridge

The Day 2 lecture introduces phase kickback. We only need its simplest instance. Since  $X|-\rangle = -|-\rangle$ , a controlled- $X$  with target  $|-\rangle$  acts as

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}}|-\rangle \mapsto \frac{|0\rangle - |1\rangle}{\sqrt{2}}|-\rangle.$$

The target ends in the same physical state, while a relative minus sign appears in the control. Apply  $H$  to the control and it becomes  $|1\rangle$ . The controlled operation has converted a property of the target into a measurable interference pattern on the control.

For the reversible function oracle above, the same reasoning gives

$$O_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle.$$

You do not need QFT or phase estimation to understand this identity. It is just the two possible actions  $\mathbb{I}|-\rangle = |-\rangle$  and  $X|-\rangle = -|-\rangle$ .

### Why this matters for LCU

Later, the ancilla will label different unitary operations. A final inverse preparation will recombine those labeled paths. It will not generally return every amplitude to ancilla 0: the part that returns is precisely the matrix action we want.

### Pause and try 4

In the garbage example, why can you not simply delete the workspace ket from the Bell state and call the remaining system  $|+\rangle$ ? Use the two different measurement predictions to explain.

**Source connection.** Lin's public notes, sections 1.7 and 2.1, provide the uncomputation and phase-kickback background. These are separate public notes, not the Duke course handouts. [Open reference.](#)

## 5. Interfering two operations: the Hadamard test

### DAY 2, TUTORIAL 1 + THE HADAMARD-TEST EXERCISE

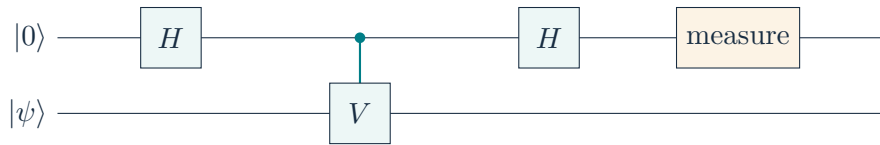
We will use an ancilla to put “do nothing” and “apply  $V$ ” into one coherent calculation.

#### Follow the state, one gate at a time

Let  $V$  be a unitary whose controlled version we can implement. Begin with  $|0\rangle|\psi\rangle$ . Apply  $H$  to the ancilla, controlled- $V$ , and then  $H$  to the ancilla again:

$$\begin{aligned} |0\rangle|\psi\rangle &\mapsto \frac{|0\rangle|\psi\rangle + |1\rangle|\psi\rangle}{\sqrt{2}} \\ &\mapsto \frac{|0\rangle|\psi\rangle + |1\rangle V|\psi\rangle}{\sqrt{2}} \\ &\mapsto |0\rangle \frac{(\mathbb{I} + V)|\psi\rangle}{2} + |1\rangle \frac{(\mathbb{I} - V)|\psi\rangle}{2}. \end{aligned}$$

The last line comes from substituting  $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$  and  $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ , then collecting the ancilla-0 and ancilla-1 terms. Notice the sum and difference of *operators*, not just numbers.



#### Derive the probability instead of memorizing it

From Lesson 3, square the norm of the ancilla-0 branch:

$$\begin{aligned} p(0) &= \frac{1}{4} \langle \psi | (\mathbb{I} + V)^\dagger (\mathbb{I} + V) | \psi \rangle \\ &= \frac{1}{4} \langle \psi | (2\mathbb{I} + V + V^\dagger) | \psi \rangle \\ &= \frac{1 + \text{Re} \langle \psi | V | \psi \rangle}{2}. \end{aligned}$$

The second line uses  $V^\dagger V = \mathbb{I}$ . The symbol  $\text{Re}$  means real part. If  $z = \langle \psi | V | \psi \rangle$ , then the two cross terms add to  $z + z^* = 2 \text{Re } z$ .

Assign the number  $+1$  to ancilla outcome 0 and  $-1$  to outcome 1. Their average over many fresh runs estimates  $p(0) - p(1) = \text{Re} \langle \psi | V | \psi \rangle$ . One shot is a sample, not an exact expectation value.

## The first block encoding has already appeared

Take  $V = Z$  and  $|\psi\rangle = (3|0\rangle + 4|1\rangle)/5$ . Then

$$\langle\psi|Z|\psi\rangle = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}, \quad p(0) = \frac{9}{25}.$$

We can also calculate the output directly:

$$\frac{\mathbb{I} + Z}{2} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{\mathbb{I} - Z}{2} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus the joint output is  $(3/5)|00\rangle + (4/5)|11\rangle$ . If the ancilla is 0, the system is  $|0\rangle$ ; if it is 1, the system is  $|1\rangle$ . The same circuit can be viewed as a statistical test or as a way to place the nonunitary operator  $(\mathbb{I} + Z)/2$  in a measurement branch.

## Two extensions you only need to recognize

To estimate the imaginary part, insert  $S^\dagger = \text{diag}(1, -i)$  on the ancilla before its final  $H$ . Its 0-branch becomes  $(\mathbb{I} - iV)|\psi\rangle/2$ , giving  $p(0) = [1 + \text{Im}\langle\psi|V|\psi\rangle]/2$ . The sign depends on using  $S^\dagger$  rather than  $S$ .

For the SWAP test, the system consists of two input registers  $|\phi\rangle|\psi\rangle$ , and  $V$  exchanges them. Their overlap with the exchanged state is

$$(\langle\phi|\langle\psi|)(|\psi\rangle|\phi\rangle) = \langle\phi|\psi\rangle\langle\psi|\phi\rangle = |\langle\phi|\psi\rangle|^2.$$

So the same Hadamard-test formula estimates a squared overlap. That is its connection to the Day 1 lecture; you can return to its applications later.

### The idea to retain

The first Hadamard creates two paths, controlled- $V$  makes the paths perform different operations, and the final Hadamard recombines them. One output branch contains a sum of operations. LCU will replace equal weights with chosen weights.

### Pause and try 5

Use  $V = X$  and input  $|0\rangle$ . Find the unnormalized ancilla-0 branch, its probability, and the conditional system state.

**Source connection.** Childs, section 16.1, treats the Hadamard test; Lin, section 3.1, includes its imaginary-part and SWAP-test variants. [Open reference.](#)

## 6. Put the desired matrix inside a unitary

### DAY 4, LECTURE 1

We can now turn the informal idea of a useful measurement branch into the definition of block encoding.

#### Why the upper-left block is the relevant one

Let  $A$  act on an  $n$ -qubit system, with  $N = 2^n$  basis states. Add one ancilla. In our ordering, the first  $N$  joint basis labels have ancilla 0 and the next  $N$  have ancilla 1. Any joint unitary can be partitioned accordingly:

$$U = \begin{pmatrix} B & C \\ D & E \end{pmatrix}.$$

Each displayed entry is now an  $N \times N$  matrix. Initializing the ancilla to 0 places the system vector in the first half of a larger vector. Ordinary multiplication gives

$$U \begin{pmatrix} \psi \\ 0 \end{pmatrix} = \begin{pmatrix} B\psi \\ D\psi \end{pmatrix} \iff U|0\rangle|\psi\rangle = |0\rangle B|\psi\rangle + |1\rangle D|\psi\rangle.$$

The upper-left block maps ancilla-0 inputs to ancilla-0 outputs. If  $B = A/\alpha$  for a positive scale  $\alpha$ , we have an exact **block encoding of  $A$** . The other blocks complete the larger matrix into a unitary.

| input ancilla 0 | input ancilla 1 |                  |
|-----------------|-----------------|------------------|
| $A/\alpha$      | $C$             | output ancilla 0 |
| $D$             | $E$             | output ancilla 1 |

#### Read the definition as three actions

The same block is written without displaying the matrix:

$$(\langle 0| \otimes \mathbb{I}) U (|0\rangle \otimes \mathbb{I}) = A/\alpha.$$

Read right to left: **attach a zero ancilla; apply the unitary; extract the zero-ancilla output**. Nothing in this definition requires a measurement to be performed. We may keep the whole computation coherent and use the block encoding as a subroutine.

**Source connection.** Lin, section 6.2 (Definition 6.1), gives the precise block-encoding definition. [Open reference.](#)

## A complete four-by-four construction

Return to  $A = \text{diag}(1, 1/2)$ . Set  $R = \text{diag}(0, \sqrt{3}/2)$  and form

$$U_{\text{direct}} = \begin{pmatrix} A & R \\ R & -A \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & \sqrt{3}/2 \\ 0 & 0 & -1 & 0 \\ 0 & \sqrt{3}/2 & 0 & -1/2 \end{pmatrix}.$$

This is our own explicit completion. Why does it work? Both  $A$  and  $R$  are real diagonal matrices, so they commute. Also  $A^2 + R^2 = \mathbb{I}$ . Consequently,

$$U_{\text{direct}}^\dagger U_{\text{direct}} = U_{\text{direct}}^2 = \begin{pmatrix} A^2 + R^2 & AR - RA \\ RA - AR & R^2 + A^2 \end{pmatrix} = \mathbb{I}_4.$$

Its upper-left block is exactly  $A$ , so its scale is  $\alpha = 1$ . This proves a valid encoding exists for this example; a small general-purpose formula alone would not prove efficient gate synthesis for a large matrix.

Now apply it to our recurring input  $|\psi\rangle = (3|0\rangle + 4|1\rangle)/5$  with ancilla 0:

$$U_{\text{direct}} \begin{pmatrix} 3/5 \\ 4/5 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3/5 \\ 2/5 \\ 0 \\ 2\sqrt{3}/5 \end{pmatrix}.$$

The success branch is  $(3|0\rangle + 2|1\rangle)/5 = A|\psi\rangle$ . Its probability is  $13/25$ . The failure branch has probability  $12/25$ . They add to one, so no probability has vanished. Conditioned on success, the system state is

$$\frac{3|0\rangle + 2|1\rangle}{\sqrt{13}}.$$

The original amplitude ratio was  $3 : 4$ ; the new ratio is  $3 : 2$ . That is the intended filtering action of  $A$ .

### There are three different objects

The unitary output is a normalized *joint* state. The selected branch is the generally unnormalized vector  $A|\psi\rangle/\alpha$ . The conditional *system* state is  $A|\psi\rangle/\|A|\psi\rangle\|$ . Do not substitute one for another.

## Scaling, accuracy, and what the definition does not promise

If the encoding is exact, the probability of obtaining all-zero ancillas is

$$p_{\text{good}} = \frac{\|A|\psi\rangle\|^2}{\alpha^2}.$$

This depends on the input. For our  $A$ , inputs  $|0\rangle$  and  $|1\rangle$  succeed with probabilities 1 and  $1/4$ , respectively. Independent retries on fresh preparations take  $1/p_{\text{good}}$  attempts on average when  $p_{\text{good}} > 0$ .

## Why a scale factor can be necessary

The **operator norm**  $\|A\|$  is the largest output length  $\|Av\|$  among vectors  $v$  of length one. A block of a unitary cannot increase an input's length beyond one: it is only part of a norm-one joint output. Therefore an exact block encoding requires  $\alpha \geq \|A\|$ .

For  $A' = \text{diag}(2, 1)$ , the largest stretch is 2. Choose  $\alpha = 2$ , and the encoded block  $A'/2$  is the same matrix as our example. If we choose an unnecessarily large scale, we reduce the amplitude and probability in the useful branch. For a fixed  $A$  and input, doubling  $\alpha$  quarters the success probability.

## The notation you will encounter in the lecture

An  $(\alpha, a, \varepsilon)$  block encoding uses  $a$  ancilla qubits and obeys

$$\|A - \alpha(|0^a\rangle\langle 0^a| \otimes \mathbb{I})U(|0^a\rangle\langle 0^a| \otimes \mathbb{I})\| \leq \varepsilon.$$

Here  $|0^a\rangle$  means all  $a$  ancillas are zero. The error is measured in operator norm. In this convention,  $\varepsilon$  bounds the error in  $A$ , not in  $A/\alpha$ . An exact encoding has  $\varepsilon = 0$ . Always check an author's convention before comparing errors.

If the actual selected block is  $B$ , its actual probability is  $\|B|\psi\rangle\|^2$ . The exact formula with  $A/\alpha$  becomes approximate. Even a small absolute error can matter for a normalized output when  $A|\psi\rangle$  is very small.

### Pause and try 6

What does  $U_{\text{direct}}$  do to  $|0\rangle_{\text{anc}}|1\rangle_{\text{sys}}$ ? State the two ancilla probabilities. Then repeat for the input  $|0\rangle_{\text{anc}}|0\rangle_{\text{sys}}$ .

**Source connection.** The scaling and approximation notation follows Lin's Definition 6.1. The displayed examples provide direct checks of its meaning. [Open reference.](#)

## 7. Build a matrix by adding unitary branches

### DAY 4, LECTURE 2: FOCUS ON GENERAL LCU

The Hadamard test added two operations equally. We now choose how much of each operation appears.

#### A weighted sum for our running matrix

The identity and  $Z$  are both unitary. Yet their weighted sum

$$A = \frac{3}{4}\mathbb{I} + \frac{1}{4}Z = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$$

is not. This is a **linear combination of unitaries**, or LCU. Applying  $\mathbb{I}$  and then  $Z$  would give the product  $Z$ , not this sum. To add operations, we use separate ancilla-labeled paths and then interfere them.

Choose a preparation unitary  $P$  such that

$$P|0\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle.$$

Why square roots? Each desired weight will appear as the product of one amplitude from preparation and another from inverse preparation. Use  $\sqrt{3/4}$  and  $\sqrt{1/4}$  so those products are  $3/4$  and  $1/4$ .

Define SELECT to apply  $\mathbb{I}$  for ancilla 0 and  $Z$  for ancilla 1. It is just controlled- $Z$  in this example. Starting from  $|0\rangle|\psi\rangle$ , the first two stages produce

$$\frac{\sqrt{3}}{2}|0\rangle|\psi\rangle + \frac{1}{2}|1\rangle Z|\psi\rangle.$$

Now apply  $P^\dagger$  to the ancilla. To find only the ancilla-0 output, use

$$\langle 0|P^\dagger = (P|0\rangle)^\dagger = \frac{\sqrt{3}}{2}\langle 0| + \frac{1}{2}\langle 1|.$$

The zero branch is therefore

$$\frac{\sqrt{3}}{2} \frac{\sqrt{3}}{2} |\psi\rangle + \frac{1}{2} \frac{1}{2} Z |\psi\rangle = \left( \frac{3}{4} \mathbb{I} + \frac{1}{4} Z \right) |\psi\rangle = A |\psi\rangle.$$

Every step is an ordinary gate operation on the larger system. The nonunitary matrix emerges as one block of their combined unitary.

**Source connection.** Arrazola, Guala, and Soni's PennyLane tutorial explains the PREPARE–SELECT–inverse-PREPARE construction. Our two-by-two example and its full matrix calculation are developed here. [Open reference.](#)

## The general construction, with every symbol explained

Suppose we have an expression

$$A = \sum_{j=0}^{m-1} c_j V_j, \quad c_j \geq 0, \quad \alpha = \sum_{j=0}^{m-1} c_j > 0.$$

The summation sign means add the  $m$  listed terms. The integer  $j$  labels a term,  $c_j$  is its weight, and each  $V_j$  is unitary on the system. With  $a = \lceil \log_2 m \rceil$  ancilla qubits we can label all terms, padding unused labels with weight zero if necessary.

**PREPARE** is a unitary  $P$  whose action on the all-zero ancilla state is

$$P|0^a\rangle = \sum_j \sqrt{\frac{c_j}{\alpha}} |j\rangle.$$

This is normalized because the squared amplitudes sum to  $\sum_j c_j/\alpha = 1$ . The choice of  $P$  on the remaining input basis states must complete a unitary, but only this prepared column is needed for the selected block.

**SELECT** preserves each label and applies its operation:

$$\text{SELECT}|j\rangle|\psi\rangle = |j\rangle V_j|\psi\rangle.$$

It must be a valid unitary on unused labels too; assigning the identity there is sufficient. Combine the three stages:

$$U = (P^\dagger \otimes \mathbb{I}) \text{SELECT}(P \otimes \mathbb{I}).$$

After the first two stages the state is  $\sum_j \sqrt{c_j/\alpha} |j\rangle V_j|\psi\rangle$ . Extracting the zero branch after  $P^\dagger$  supplies another factor  $\sqrt{c_j/\alpha}$  for each term. Thus

$$(\langle 0^a| \otimes \mathbb{I}) U (|0^a\rangle \otimes \mathbb{I}) = \sum_j \frac{c_j}{\alpha} V_j = \frac{A}{\alpha}.$$

### What inverse preparation actually does

It recombines the term labels in a chosen basis. It does not usually clean the ancillas to zero with certainty. The all-zero component is the desired weighted sum, and other components preserve the rest of the probability.

**Source connection.** The general LCU construction appears in the PennyLane tutorial and in Lin's section 7.3. [Open reference.](#)

## Signs, costs, and composing encodings

### Negative or complex coefficients are allowed

If a coefficient is  $c_j = |c_j|e^{i\theta_j}$ , absorb its phase into  $W_j = e^{i\theta_j}V_j$ . This is still unitary. Prepare using the positive magnitudes  $|c_j|$  and set  $\alpha = \sum_j |c_j|$ . SELECT must implement the phased  $W_j$ , because branch-relative phases matter.

For example,  $2\mathbb{I} - Z$  has positive weights 2 and 1 attached to operations  $\mathbb{I}$  and  $-Z$ . The LCU scale is 3. Preparing with coefficients 2 and  $-1$  directly as probabilities would be invalid.

### What controls the expense?

The term count  $m$  determines the size of the label register, but does not by itself determine the gate count. You must also construct  $P$ , construct SELECT, and account for the scale  $\alpha$ . Two decompositions of the same matrix can have very different costs and scales. A short classical formula helps only when its ingredients have useful quantum circuits.

### A useful warning about matrix products

Suppose  $U_A$  and  $U_B$  encode  $A/\alpha$  and  $B/\beta$ . Their adjoints encode the corresponding adjoint blocks. To encode the product  $AB/(\alpha\beta)$  by sequential application, one valid construction uses *separate ancilla registers*: apply  $U_B$  first, then  $U_A$ , and extract zero in both registers. A failed first register cannot secretly re-enter the all-zero outcome through an operation on the second register.

Reusing the same ancilla without a construction that controls this leakage is generally wrong. Block multiplication gives

$$\begin{pmatrix} A & C \\ D & E \end{pmatrix}_{00}^2 = A^2 + CD,$$

where the subscript means the upper-left block. In our direct encoding,  $U_{\text{direct}}^2 = \mathbb{I}$ , even though  $A^2 = \text{diag}(1, 1/4) \neq \mathbb{I}$ . The extra path through the other ancilla sector changes the result. This is one reason later matrix-function algorithms require carefully designed sequences.

#### Pause and try 7

Find a positive-weight LCU for  $B = \text{diag}(1, 1/3)$  using  $\mathbb{I}$  and  $Z$ . What ancilla state should PREPARE produce? What is the success probability on system input  $|1\rangle$ ?

**Source connection.** Gilyén, Su, Low, and Wiebe give formal constructions for linear combinations and products of block encodings. [Open reference.](#)

## 8. Work through the entire LCU circuit

### DAY 4, TUTORIAL 3 + THE LCU-CIRCUIT EXERCISE

A construction is understood when you can verify its full unitary, its selected block, and its output probabilities.

#### Make PREPARE and SELECT explicit

For our recurring matrix, choose

$$P = \begin{pmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix}, \quad \text{SELECT} = \text{diag}(1, 1, 1, -1).$$

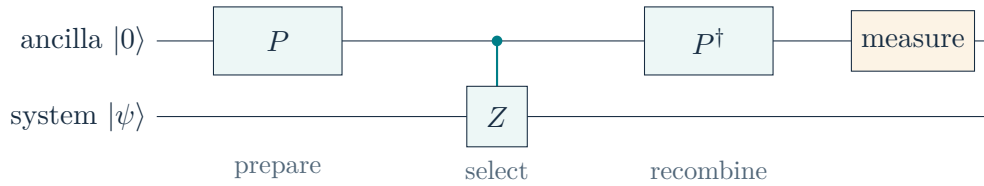
The first column of  $P$  is the state we wanted. Its two columns have length one and zero inner product, so it is unitary. It is also a rotation  $R_y(\pi/3)$  in the convention

$$R_y(\theta) = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}.$$

SELECT is controlled-Z, often written CZ. Multiplying the three matrices gives

$$U_{\text{LCU}} = (P^\dagger \otimes \mathbb{I}) \text{SELECT} (P \otimes \mathbb{I}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & -\sqrt{3}/2 \\ 0 & 0 & 1 & 0 \\ 0 & -\sqrt{3}/2 & 0 & -1/2 \end{pmatrix}.$$

For example, its upper-left block is  $(3/4)\mathbb{I} + (1/4)Z$ . Its upper-right block is  $(\sqrt{3}/4)(Z - \mathbb{I})$ , explaining the negative off-block entry.



This unitary differs from  $U_{\text{direct}}$  outside the desired block. Both encode the same  $A$  with the same scale and therefore have identical success probabilities on every system input. The definition specifies the useful block, not a unique completion.

## A numerical check you can reproduce

You do not need programming to follow the guide. If you use Python with NumPy, the following independently written calculation checks the exact example. Here `@` is matrix multiplication, `kron` builds a tensor product, and `conj().T` takes a conjugate transpose.

```
import numpy as np

I = np.eye(2)
A = np.diag([1.0, 0.5])
P = np.array([[np.sqrt(3)/2, -0.5],
              [0.5, np.sqrt(3)/2]])
SELECT = np.diag([1.0, 1.0, 1.0, -1.0])
PREP = np.kron(P, I)
U = PREP.conj().T @ SELECT @ PREP

assert np.allclose(U.conj().T @ U, np.eye(4))
assert np.allclose(U[:2, :2], A)

psi = np.array([3.0, 4.0]) / 5
initial = np.kron([1.0, 0.0], psi)
output = U @ initial
good = output[:2] # ancilla 0, in our ordering
p_good = np.vdot(good, good).real
conditional = good / np.sqrt(p_good)

print(output.round(6))
print(round(p_good, 6))
print(conditional.round(6))
```

The expected output is

$$\begin{bmatrix} 0.6 & 0.4 & 0.0 & -0.692820 \\ & & 0.52 & \\ & & & 0.832050 & 0.554700 \end{bmatrix}.$$

The minus sign in the failure branch differs from the direct construction. The success branch is still  $(3, 2)^T/5$ , with probability  $13/25$  and normalized output  $(3, 2)^T/\sqrt{13}$ .

## What to do in the PennyLane tutorial

Implement the same preparation, controlled selection, and inverse preparation. Inspect the full circuit matrix with an explicit ancilla-first wire order, then compare its selected block to the intended  $A/\alpha$ . A simulator's state vector is useful for debugging; a physical measurement does not reveal that vector in one shot.

**Resource interpretation.** This example uses two qubits, two single-qubit rotations, and one CZ before measurement, if those rotations and CZ count as available gates. A fault-tolerant gate library may need many elementary gates to approximate the rotations to a chosen accuracy. Dense simulator matrices also consume memory exponential in the qubit count.

### Pause and try 8

Change  $\psi$  to  $|+\rangle = (1, 1)^T/\sqrt{2}$ . Predict the success probability and conditional output before running the calculation.

## 9. Exploit the structure of the matrix

### DAY 5, LECTURE 3

*The next question is no longer whether a block encoding exists. It is how to build one with operations whose cost we understand.*

#### Structured example 1: diagonal filtering

Suppose  $D = \text{diag}(d_0, \dots, d_{N-1})$ , with real  $d_j \in [0, 1]$ . For each system basis label  $j$ , rotate an ancilla by an angle satisfying

$$\theta_j = 2 \arccos(d_j).$$

Because  $\cos(\theta_j/2) = d_j$  and  $\sin(\theta_j/2) = \sqrt{1 - d_j^2}$ , this gives

$$|0\rangle|j\rangle \mapsto (d_j|0\rangle + \sqrt{1 - d_j^2}|1\rangle)|j\rangle.$$

On input  $|\psi\rangle = \sum_j b_j|j\rangle$ , linearity produces a zero-ancilla branch  $\sum_j d_j b_j|j\rangle = D|\psi\rangle$ . This is an exact block encoding with scale one. It is unitary because each fixed- $j$  operation is a complete ancilla rotation and the label  $j$  is preserved.

For our example, the angles are  $\theta_0 = 0$  and  $\theta_1 = 2\pi/3$ . A rotation on the ancilla controlled by system value 1 therefore provides another encoding of  $A$ . Again, the unspecified blocks can differ while the useful block agrees.

#### From a formula to an access circuit

If  $d_j$  has a simple formula, a circuit may calculate its value into temporary workspace, use it to set the rotation, then uncompute the workspace. The arithmetic and angle approximation have a cost. If instead the  $d_j$  are an arbitrary list of  $N$  unrelated numbers, loading that list can be expensive. One must specify the access method before claiming a speedup.

#### What “sparse access” is telling you

A sparse matrix has relatively few nonzero entries in each row or column. Sparse-access constructions typically assume coherent procedures that locate nonzero entries and return their values. Having a sparse matrix on paper is not the same thing as having inexpensive quantum procedures for those tasks.

The matrix-access lecture and reference explore this route. For a first pass, identify the value oracle, the index oracle, their workspace, and the resulting normalization. You can postpone the detailed synthesis of a general sparse-access encoding until you can check the diagonal construction above.

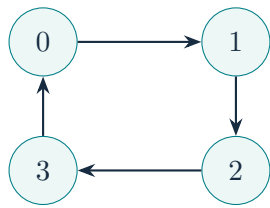
**Source connection.** Soni, Guala, and Jahangiri’s tutorial introduces matrix-entry amplitude oracles and structured/sparse circuit constructions. The diagonal rotation example here is derived directly. [Open reference.](#)

## Structured example 2: neighboring points on a ring

Take four positions labeled 0, 1, 2, 3. Define a shift  $S|j\rangle = |j + 1 \bmod 4\rangle$ . It is unitary: it permutes the basis labels without losing information. Its inverse  $S^\dagger$  shifts the other way. The matrix

$$L = 2\mathbb{I} - S - S^\dagger = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

compares each value with its two neighbors. This is a discrete Laplacian on a four-point ring. The example is built here to connect LCU to structured matrices; it is not presented as an example taken from Chao Yang's recording.



$S$  moves one step around the ring.  
 $S^\dagger$  moves one step back.

This is an LCU with weights 2, 1, 1 and unitaries  $\mathbb{I}, -S, -S^\dagger$ . Use two ancilla qubits, scale  $\alpha = 4$ , and prepare

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{1}{2}|10\rangle.$$

SELECT applies the three corresponding operations; the unused label 11 may apply  $\mathbb{I}$ . Inverse preparation gives a selected block  $L/4$ . The minus signs are phases on selected branches and are part of the circuit.

On system input  $|0\rangle$ , the selected branch is  $(2|0\rangle - |1\rangle - |3\rangle)/4$ . Its probability is  $(4 + 1 + 1)/16 = 3/8$ . On the uniform state  $(|0\rangle + |1\rangle + |2\rangle + |3\rangle)/2$ , the selected branch is zero because both shifts leave that state unchanged. Success is then impossible: the matrix annihilates that input.

### What to listen for in the structured-matrix lecture

Ask how structure becomes gates: Which operations are permutations? Which amplitudes are loaded? What workspace must be reversed? What scale results? How does the cost grow with  $n = \log_2 N$ ? These questions turn the definition into an algorithm.

## 10. A first look beyond block encoding

### THE REST OF DAY 4 LECTURE 2; THEN DAY 5 LECTURE 4

*You have reached the main goal. This short extension explains what the next lectures are trying to achieve without assuming their full machinery.*

#### Singular values describe how a matrix stretches directions

For a general matrix, the singular value decomposition writes

$$A = W\Sigma V^\dagger.$$

Here  $V$  describes orthonormal input directions,  $W$  describes orthonormal output directions, and  $\Sigma$  contains nonnegative stretch factors  $\sigma_j$ , the **singular values**. Equivalently,  $A|v_j\rangle = \sigma_j|w_j\rangle$ . The largest  $\sigma_j$  is the operator norm. For our positive diagonal matrix, the directions are the usual basis and the stretch factors are 1 and  $1/2$ .

An eigenvector, by comparison, obeys  $A|v\rangle = \lambda|v\rangle$ : its input and output directions are the same. A Hermitian matrix satisfies  $A = A^\dagger$  and has real eigenvalues. These notions agree in simple positive diagonal examples but should not be interchanged for an arbitrary matrix.

**Source connection.** For the linear algebra behind singular values, see Gilbert Strang's MIT lecture. [Open reference.](#)

#### QSP and QSVT turn a scalar function into a matrix operation

Quantum signal processing (QSP) designs phased unitary sequences whose selected amplitudes are polynomials in a scalar signal. Quantum singular value transformation (QSVT) uses related sequences built from a block encoding, its adjoint, and phases associated with selected subspaces to transform singular values.

The input signal is normalized: its singular values are  $\sigma_j/\alpha$ . For an appropriate odd polynomial  $p$ , the transformed operator has the singular-vector form  $Wp(\Sigma/\alpha)V^\dagger$ . Even-parity constructions act within a corresponding input or output singular-vector space instead. This is not entry-by-entry application of  $p$  to  $A$ , nor is it ordinary  $p(A)$  for a general non-Hermitian matrix.

For the positive diagonal example, the distinction is easy to see numerically. A transformation corresponding to  $p(x) = x^3$  changes diagonal entries  $1, 1/2$  to  $1, 1/8$ . A reciprocal-like transformation would be useful for solving a linear system, but  $1/x$  diverges near zero and exceeds one on part of  $(0, 1]$ . One must restrict the spectral domain away from zero, rescale, and approximate with an admissible bounded polynomial. Parity and implementation conditions also matter.

**Source connection.** PennyLane's introductory QSVT tutorial gives a visual entry point. The formal conditions are in Gilyén et al., cited in the references. [Open reference.](#)

## Success amplification is a separate operation

Suppose a preparation circuit produces

$$\sqrt{p}|\text{good}\rangle + \sqrt{1-p}|\text{bad}\rangle,$$

where the two displayed states are normalized and orthogonal. Write  $\sqrt{p} = \sin \theta$ . Under the usual amplitude-amplification assumptions, alternating two reflections rotates the state in this two-dimensional plane. After  $k$  rounds the good probability has the form  $\sin^2((2k+1)\theta)$ . A **reflection** is a unitary sign flip on a chosen subspace, leaving its orthogonal complement unchanged.

With suitable access to the preparation and success reflections and appropriate choice of the number of rounds, constant success can be reached using order  $1/\sqrt{p}$  calls, instead of order  $1/p$  independent retries. Unknown success probabilities require additional care; repeatedly applying the same round blindly can rotate past the target.

**Source connection.** PennyLane’s amplitude-amplification tutorial explains the reflection geometry and its assumptions. [Open reference.](#)

## Why the word “oblivious” appears

Standard amplitude amplification often uses a reflection involving a particular prepared input state. **Oblivious amplitude amplification** instead exploits a subroutine’s structure so its reflections can act on the ancilla success subspace without needing a reflection about an unknown system input. The exact and approximate versions have conditions on the encoded operation.

Do not interpret this as a universal procedure that deterministically implements every normalized nonunitary map. Our matrix  $A$  has different success amplitudes on  $|0\rangle$  and  $|1\rangle$ . Treating it like a uniformly scaled unitary would ignore that difference. QSVT provides a broader framework for manipulating singular values, but its transformations and approximation domains still have to be specified.

Hamiltonian simulation is another application: given a Hermitian matrix  $K$  representing a Hamiltonian, the desired evolution is  $e^{-itK}$  in units where  $\hbar = 1$ . The exponential is a matrix function. You can study its approximation after you understand how  $K$  is accessed and encoded; it is not a prerequisite for the constructions above.

### A sensible stopping point

You understand basic block encoding when you can derive the selected block, compute its success probability for a chosen input, and explain the cost of the circuit’s ingredients. QSVT, oblivious amplitude amplification, and Hamiltonian simulation build on that foundation.

**Source connection.** Gilyén et al. develop singular value transformation, including robust oblivious amplitude amplification and quantum matrix-arithmetic applications. [Open reference.](#)

## Worked answers

### CHECK YOUR REASONING

*If an answer feels mysterious, revisit the indicated lesson before adding more machinery.*

#### 1. Gate order and probabilities

$X|0\rangle = |1\rangle$ , then  $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$ . The amplitudes are  $1/\sqrt{2}, -1/\sqrt{2}$  and both probabilities are  $1/2$ . Squaring happens after the gates, not between them.

#### 2. A controlled gate on a superposition

After  $H$  on the ancilla, the state is  $(|01\rangle + |11\rangle)/\sqrt{2}$ . Controlled- $X$  leaves  $|01\rangle$  alone and changes  $|11\rangle$  to  $|10\rangle$ . The result is  $(|01\rangle + |10\rangle)/\sqrt{2}$ , or  $(0, 1/\sqrt{2}, 1/\sqrt{2}, 0)^T$  in our order.

#### 3. Branch first, normalize second

The ancilla-0 branch is  $(2|0\rangle + |1\rangle)/3$ . Its squared norm is  $5/9$ . Divide by  $\sqrt{5}/3$  to get the conditional state  $(2|0\rangle + |1\rangle)/\sqrt{5}$ . The remaining ancilla outcome has probability  $4/9$  and conditional system state  $|1\rangle$ .

#### 4. Why workspace cannot just be deleted

If you replaced the system in the entangled state by  $|+\rangle$ , you would predict that a subsequent  $H$  produces 0 with certainty. The joint-state calculation gives probability  $1/2$  instead. Workspace labels preserve information about the branches, so ignoring the register is not the same operation as coherently uncomputing it.

#### 5. Hadamard-test postselection

The good branch is  $(\mathbb{I} + X)|0\rangle/2 = (|0\rangle + |1\rangle)/2$ . Its probability is  $1/4 + 1/4 = 1/2$ . Dividing by  $1/\sqrt{2}$  gives  $|+\rangle$ . The state  $(|0\rangle + |1\rangle)/2$  is a branch vector, not a normalized qubit state.

#### 6. Direct block encoding on the basis states

On  $|01\rangle$ , the output is  $\frac{1}{2}|01\rangle + \frac{\sqrt{3}}{2}|11\rangle$ . The ancilla probabilities are  $1/4$  and  $3/4$ ; either conditional system state is  $|1\rangle$ . On  $|00\rangle$ , the output is  $|00\rangle$  and ancilla success has probability one. Equal conditional output directions need not mean equal success probabilities.

## Answers 7–8 and a final self-check

### 7. Solve for the LCU weights

Write  $B = u\mathbb{I} + vZ$ . Matching its two diagonal entries gives  $u + v = 1$  and  $u - v = 1/3$ . Add these equations:  $2u = 4/3$ , so  $u = 2/3$  and  $v = 1/3$ . The scale is  $\alpha = u + v = 1$ . Prepare the ancilla state

$$\sqrt{2/3}|0\rangle + \sqrt{1/3}|1\rangle.$$

On system input  $|1\rangle$ , the selected branch is  $\frac{1}{3}|1\rangle$ . The success probability is  $1/9$  and, conditional on success, the system is  $|1\rangle$ .

### 8. A new input to the same circuit

For  $|\psi\rangle = |+\rangle$ , the selected branch is

$$A|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{2\sqrt{2}}|1\rangle.$$

Its squared norm is  $1/2 + 1/8 = 5/8$ . Normalization gives  $(2|0\rangle + |1\rangle)/\sqrt{5}$ . The circuit is unchanged; the success probability changes with the input.

## The whole story in one calculation

Suppose someone hands you a circuit  $U$  and says it encodes  $A$ . Work through these questions:

1. **Which qubits are the system and which are ancillas?** State the basis ordering and the initial ancilla state explicitly.
2. **What is the selected block?** Calculate  $(\langle 0^a| \otimes \mathbb{I})U(|0^a\rangle \otimes \mathbb{I})$  and compare it with  $A/\alpha$ .
3. **Is the full circuit unitary?** A correct-looking upper-left matrix by itself is insufficient.
4. **What happens to an input?** Compute the unnormalized selected branch, square its norm, then normalize if that norm is nonzero.
5. **How is the circuit built?** Identify preparation, controlled operations, matrix-access assumptions, workspace, and precision costs.

For our LCU circuit, these answers are: one ancilla before one system qubit; upper-left block  $A$  and scale one; a product of three unitaries; success probability  $\|A|\psi\rangle\|^2$ ; and the explicit gates  $P$ ,  $CZ$ , and  $P^\dagger$ .

### Explain it aloud

“I initialize extra qubits, apply a reversible operation to the larger system, and identify the output component where those extra qubits are zero. That component contains my matrix acting on the input, divided by a known scale. Its squared length determines the chance of selecting it.”

## Notation you can look up while reading

### QUICK REFERENCE

*Every symbol below was introduced in the lessons. This page is for returning to them without breaking the flow.*

| Notation                       | Meaning                                                                                 |
|--------------------------------|-----------------------------------------------------------------------------------------|
| $ \psi\rangle$                 | A column vector representing a pure state when normalized.                              |
| $\langle\psi $                 | The conjugate-transposed row vector.                                                    |
| $\langle\phi \psi\rangle$      | An inner product: multiply the bra by the ket to get a number.                          |
| $ \phi\rangle\langle\psi $     | An outer product: multiply a column by a row to get a matrix.                           |
| $ z ^2$                        | Squared magnitude of a complex number; if $z = x + iy$ , this is $x^2 + y^2$ .          |
| $\ v\ $                        | Vector length, the square root of the sum of squared magnitudes.                        |
| $\ A\ $                        | Operator norm: the largest stretch of a unit vector by $A$ .                            |
| $A^\dagger$                    | Conjugate transpose; equals the inverse when $A$ is unitary.                            |
| $\mathbb{I}$ or $\mathbb{I}_N$ | Identity matrix, with the dimension specified when helpful.                             |
| $\otimes$                      | Tensor product; combines registers or their separate operations.                        |
| $ 0^a\rangle$                  | The all-zero state of $a$ ancilla qubits.                                               |
| $\Pi_0$                        | Projector onto the all-zero ancilla subspace, leaving the system intact.                |
| $\alpha$                       | Positive scale for an encoding of $A$ ; the selected block is $A/\alpha$ when exact.    |
| $\varepsilon$                  | Allowed operator-norm error in the stated convention.                                   |
| $P, P^\dagger$                 | Ancilla preparation and its inverse.                                                    |
| $V_j, \text{SELECT}$           | Unitary for term $j$ ; the coherent operation that selects it using the label register. |
| $\sigma_j$                     | Singular value, a nonnegative stretch factor.                                           |

**Convention reminder.** A circuit executes left to right; an operator product acts right to left. A ket with two digits uses ancilla first in this guide. Some sources use different letters for the number of ancillas or for the normalization scale.

## References and a guide to further reading

### PRIMARY SOURCES CONSULTED

*Links are clickable. Consult the indicated sections rather than reading every source from the beginning.*

1. **Duke summer-school program.** [Program & Lecturers, 2026](#) [Gene Golub SIAM Summer School](#). Used for the lecture mapping, together with the text you supplied. Follow this page to the recordings and authorized course resources.
2. **John Watrous / IBM Quantum Learning.** [Single systems: Quantum information](#). State vectors, amplitudes, the Born rule, and unitary operations. Best companion for Lesson 1 and the beginning of Lesson 2.
3. **John Watrous / IBM Quantum Learning.** [Multiple systems: Quantum information](#). Tensor products, entanglement, partial measurements, and controlled operations. Best companion for Lessons 2–3.
4. **John Watrous / IBM Quantum Learning.** [Quantum circuits: Circuits](#), and [Limitations on quantum information](#). Circuit reading, global phase, and no cloning.
5. **Gilbert Strang / MIT OpenCourseWare.** [18.06SC: Multiplication and Inverse Matrices](#). Extra support if the row-by-column calculations are unfamiliar. The goal here is fluency with two-by-two matrices, not completion of an entire linear algebra course.
6. **Andrew M. Childs.** [Lecture Notes on Quantum Algorithms](#), 17 April 2025. Sections 1.2–1.4: circuits, universality, reversibility. Section 16.1: the Hadamard test. A broader and more advanced reference than the present guide.
7. **Lin Lin.** [Lecture Notes on Quantum Algorithms for Scientific Computation](#), 3 March 2022. Section 1.7: uncomputation; section 3.1: Hadamard and SWAP tests; section 6.2: block encoding; section 7.3: LCU; Chapter 8: singular value transformation. In this version, the main block-encoding discussion begins at printed page 80, and Definition 6.1 spans printed pages 81–82. The PDF contains 132 pages.
8. **Juan Miguel Arrazola, Diego Guala, and Jay Soni / PennyLane.** [Linear combination of unitaries and block encodings](#). Published 24 October 2023; page reports an update on 27 May 2026. The PREPARE and SELECT discussion is a useful companion to Lesson 7; its implementation examples support Lesson 8.

## References continued and source-use notes

9. **Jay Soni, Diego Guala, and Soran Jahangiri / PennyLane.** [Block encoding with matrix access oracles](#). Published 27 November 2023; page reports an update on 15 June 2026. Read the opening oracle definitions, then the structured- and sparse-matrix sections after Lesson 9. Rotation-angle conventions should always be checked against the gate's actual definition.
10. **Gilbert Strang / MIT OpenCourseWare.** [18.06SC, Lecture 29: Singular Value Decomposition](#). The input-direction, stretch, and output-direction interpretation supports the final extension lesson.
11. **PennyLane.** [Intro to QSVT](#). An entry point to the polynomial viewpoint. Read after the main block-encoding construction; the polynomial conditions and circuit conventions are additional material, not assumed prerequisites here.
12. **PennyLane.** [Intro to Amplitude Amplification](#). Reflection geometry and the effect of repeated amplification rounds. Useful before attempting the amplitude-amplification parts of the Day 4 exercise and Day 5 lecture.
13. **András Gilyén, Yuan Su, Guang Hao Low, and Nathan Wiebe.** [Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics](#), arXiv:1806.01838. Primary research source for QSVT, block-encoding constructions, and robust oblivious amplitude amplification. This is a technical reference to return to later.

## How the sources were used

The source connections identify the references for the concepts and the places to continue reading. The running calculations, the ring example, the exercises, the numerical verification, and all figures were composed for this guide. The explanations are a pedagogical synthesis of standard mathematics, rather than extracts from lecture transcripts.

The figures are original vector drawings so their labels and register order match the text. This document includes links to external notes and tutorials rather than reproducing their pages or video frames. It contains no screenshots from the Duke course materials. No particular statement here should be interpreted as a quotation from an unreviewed recording.

## Your next session

Begin with Lesson 1 and Day 1 Lecture 1. Work through the gate example until you can calculate both output amplitudes yourself. Then move to Lecture 2. The most important checkpoint before Day 4 is Lesson 3: if you can extract a branch, calculate its probability, and normalize it, the block-encoding definition will have a concrete meaning.